

Topology & Multivariable Calculus

FMA_2F002_EP

Lecture Notes

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1 Norms in Vector Spaces

1.1 Setting

Let $E, F, G, \dots$ be vector spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

- If $\mathbb{K} = \mathbb{R}$, for $x \in \mathbb{R}$, $|x| = \max(x, -x)$.
- If $\mathbb{K} = \mathbb{C}$, for $z = x + iy$, we have $|z| = \sqrt{x^2 + y^2} = l$. In polar form, $z = le^{i\theta}$ where $l \geq 0$ and $\theta \in [0, 2\pi)$.

1.2 Norms

Definition 1: Norm

A norm on E is a map $N : E \rightarrow \mathbb{R}_{\geq 0} = \mathbb{R}^+ = [0, +\infty)$ satisfying the following 3 properties:

1. **Triangular inequality:** $\forall x, y \in E, N(x + y) \leq N(x) + N(y)$.
2. **Absolute homogeneity:** $\forall \lambda \in \mathbb{K}, \forall x \in E, N(\lambda x) = |\lambda|N(x)$.
3. **Positive definiteness:** $N(x) = 0 \Leftrightarrow x = 0$.

Remark 2

Positive definiteness is equivalent to the fact that N "separates points". That is, $\forall x, y \in E$, $N(x - y) = 0$ if and only if $x = y$.

Proposition 3: Reverse Triangular Inequality

The triangular inequality is equivalent to the reverse triangular inequality: $\forall x, y \in E$, $|N(x) - N(y)| \leq N(x - y)$.

PROOF. Let $x, y \in E$. Then $N(x) = N(x - y + y) \leq N(x - y) + N(y)$, which implies $N(x) - N(y) \leq N(x - y)$. Similarly, $N(y) - N(x) \leq N(y - x)$. Since $N(x - y) = |-1|N(y - x)$, we obtain $|N(x) - N(y)| \leq N(x - y)$. $\square$

Example 4: Common Norms

- $|\cdot|$ is a norm on $\mathbb{R}$.
- On $E = \mathbb{R}^d$, Sup Norm $\|x\|_{\infty} = \max_{i=1}^d |x_i|$.
- Euclidean norm on $\mathbb{R}^d$: $\|x\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$.
- On $F = C^0([0, 1], \mathbb{R})$, Sup Norm $\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|$.
- On F , L_1 norm $\|f\|_{L^1} = \int_0^1 |f(x)| dx$.
- On F , L_2 norm $\|f\|_{L^2} = \sqrt{\int_0^1 |f(x)|^2 dx}$.

Lemma 5: Norms on $\mathbb{R}$

Let $N : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be a norm. Then $\exists \alpha > 0$ such that $N(\cdot) = \alpha |\cdot|$.

PROOF. Let $x \in \mathbb{R}$. Then $N(x) = N(x \cdot 1) = |x|N(1)$. Let $\alpha = N(1) > 0$. $\square$

Lemma 6: Euclidean Norm on $\mathbb{R}^d$

$\|\cdot\|_2$ is a norm on $E = \mathbb{R}^d$.

PROOF. Absolute homogeneity is obvious. For positive definiteness: $\|x\|_2 = 0 \Rightarrow \sum_{i=1}^d x_i^2 = 0 \Rightarrow x_i = 0$ for all $i = 1, \dots, d \Rightarrow x = 0_E$. For the triangular inequality, we use the Euclidean scalar product $\langle x, y \rangle = \sum_{i=1}^d x_i y_i$. This product is symmetric, $\mathbb{R}$ -bilinear, and $\langle x, x \rangle = 0 \Rightarrow x = 0_E$. We have $\|x + y\|_2 = \sqrt{\langle x + y, x + y \rangle} = \sqrt{\|x\|_2^2 + \|y\|_2^2 + 2\langle x, y \rangle}$. By the Cauchy-Schwarz inequality, $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$. Thus, $\|x + y\|_2 \leq \sqrt{\|x\|_2^2 + 2\|x\|_2 \|y\|_2 + \|y\|_2^2} = \|x\|_2 + \|y\|_2$. $\square$

Lemma 7: Sup Norm on $C^0([0, 1], \mathbb{R})$

$\|\cdot\|_\infty$ is a norm on $F = C^0([0, 1], \mathbb{R})$.

PROOF. For any $f \in F$, f is bounded. Positive homogeneity: $\|\lambda f\|_\infty = \sup |\lambda f(x)| = |\lambda| \|f\|_\infty$. Positive definiteness: If $\|f\|_\infty = 0$, then $0 \leq |f(x)| \leq \sup |f(x)| = 0$, which implies $f(x) = 0$ for all x , so $f = 0_F$. Triangular inequality: $|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty$. Taking the supremum yields the result. $\square$

1.3 Distances

Definition 8: Distance

A distance on a set $X \neq \emptyset$ is a map $d : X^2 \rightarrow \mathbb{R}^+$ satisfying:

1. **Triangular inequality:** $\forall x, y, z \in X, d(x, z) \leq d(x, y) + d(y, z)$.
2. **Symmetry:** $\forall x, y \in X, d(x, y) = d(y, x)$.
3. **Positive definiteness:** $\forall x, y \in X, d(x, y) = 0 \Leftrightarrow x = y$.

Example 9: Discrete Distance

Let $X \neq \emptyset$. The discrete distance is defined as $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.

Proposition 10: Distance Induced by a Norm

Let (E, N) be a normed vector space (NVS) over $\mathbb{K}$. The map $d_N : E^2 \rightarrow \mathbb{R}^+$ given by $d(x, y) = N(x - y)$ is a distance on E .

PROOF. Symmetry: $d(x, y) = N(x - y) = N(y - x) = d(y, x)$. Positive definiteness: $d(x, y) = 0 \Leftrightarrow N(x - y) = 0 \Rightarrow x - y = 0 \Rightarrow x = y$. Triangular inequality: $d(x, z) = N(x - z) =$

$$N(x - y + y - z) \leq N(x - y) + N(y - z) = d(x, y) + d(y, z). \quad \square$$

1.4 Balls and Convexity

Definition 11: Balls

Let (X, d) be a metric space. For $x_0 \in X$ and $r \in \mathbb{R}^+$, we define:

- **Open ball:** $B(x_0, r) = \{x \in X \mid d(x_0, x) < r\}$.
- **Closed ball:** $\overline{B}(x_0, r) = \{x \in X \mid d(x_0, x) \leq r\}$.

In E , $B(0, 1)$ is defined as the unit ball.

Definition 12: Segments and Convexity

For $x, y \in E$, the closed segment is $[x, y] = \{tx + (1 - t)y \mid t \in [0, 1]\}$. The open segment is $(x, y) = [x, y] \setminus \{x, y\}$. A subset $C \subseteq E$ is **convex** if for all $x, y \in C$, the segment $[x, y] \subseteq C$.

Proposition 13: Convex Subsets

1. Convex subsets of $\mathbb{R}$ are intervals.
2. Balls in a NVS are convex.

PROOF. We prove (2) for $B = B(x_0, r)$ with $r > 0$. Let $x, y \in B$, meaning $N(x - x_0) < r$ and $N(y - x_0) < r$. Let $z = tx + (1 - t)y \in [x, y]$ with $t \in [0, 1]$. We have $d(z, x_0) = N(z - x_0) = N(t(x - x_0) + (1 - t)(y - x_0))$. By the triangular inequality, $d(z, x_0) \leq tN(x - x_0) + (1 - t)N(y - x_0) < tr + (1 - t)r = r$. Thus, $z \in B$, which implies $[x, y] \subseteq B$, meaning B is convex. $\square$

2 Sequences

Definition 1: Convergence of Sequences

Let $\underline{x} = (x_n)_{n \in \mathbb{N}}$ be a sequence in a $\mathbb{K}$ -NVS E . We say $x_n \rightarrow x_\infty$ in E (wrt N) if $N(x_n - x_\infty) \rightarrow 0$ in $\mathbb{R}$. We denote this as $x_n \rightarrow x_\infty$ or $\lim x_n = x_\infty$. If a sequence does not converge, it diverges.

Example 2: Convergence in Different Norms

Consider $E = C^0([0, 1], \mathbb{R})$ and the sequence of functions $f_n(x) = x^n$.

- For the sup norm $\|\cdot\|_\infty$, we have $\|f_n\|_\infty = 1$ for all n , so it does not converge to 0_E .
- For the L^1 norm, we have $\|f_n\|_{L^1} = \int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0$. Thus $f_n \rightarrow 0_E$ for $\|\cdot\|_{L^1}$.